In this lab we were tasked with assembling a sensor piece by piece and applying it to an apparatus. To begin, we were given an Arduino ESP 32 microcontroller and became familiar with its use by using Copilot to help make the Arduino's LED blink. After this base demo, we moved on to breadboarding, where we caused an LED to light, and demonstrated the purposes of circuit components by overloading a resistor. The resistor burnt out, as it was intended to, but our LED also stopped working, showing that parts do not always fail as they should.

After this introduction to our sensor components, we created a circuit and program (using Chat GPT) to move a servo motor. From there, we created another circuit and program to take measures of resistance from a moisture sensor and transform that data using a Chat GPT program. After practicing practical uses of these sensors and microcontroller, we began to create a more complex circuit. The goal was to make an alternate version of a sensor to determine if a window was open or closed, and to trigger an alarm if opened. We decided to use a light sensor to detect if the window was opened (the window being closed would darken the photoresistor), a button switch to arm and disarm the system, and an LED that acted as the alarm, flashing when the alarm was tripped. We attached the photoresistor, button, and LED to the Arduino, and asked Chat GPT to write the code. After a few tries we got both the code and the entire system to work!

This module gave me a great introduction to circuitry, programming, and sensors, which I will take with me throughout my engineering career. One of few difficulties arose from using AI to code. The model would often misunderstand the directions, producing logic errors, but after a few tweaks to the prompts, it would create code to perfectly match our needs. My group was able to overcome any problem that arose, completing the project very efficiently. This project allowed me to build my collaboration, programming, electronics, and my overall engineering skills.